

Per-unit Scheme for Integrated Solution of Hybrid Power-water Flow Problem

Xia Zhao, Hong Tan, Luo Wang, and Mingyi Sun

Abstract—When applying the widely-used Newton-Raphson (N-R) method to the integrated solution of the hybrid power-water flow problem of the power-water integrated energy system (PW-IES), it is found that there are numerical differences among Jacobian elements, which may yield an ill-conditioned Jacobian matrix and even cause convergence problem. Therefore, this letter proposes to per-unitize the water distribution network (WDN) to reduce the numerical differences and thus to improve the Jacobian matrix conditioning and the integrated N-R solution to the hybrid power-water flow problem.

Index Terms—Integrated energy system, load flow, Newton-Raphson method, water distribution network, power-water flow.

I. INTRODUCTION

RECENT years, there are significant efforts worldwide to develop the integrated energy system (IES) or multi-energy system (MES) toward a more efficient, flexible, and decarbonized energy system [1]. The IES seeks potential synergies among electricity, natural gas, heating, and cooling networks, and therefore suggests to plan and operate various energy vectors from a more integrated perspective. In this context, there is a growing interest in the interdependences between the power and water as well as the coordination of the power system and water distribution network (WDN).

Load flow analysis, also termed as energy flow or multi-energy flow in the field of the IES, is the basic tool for IES planning and operation. A lot of studies have been devoted to the load flow modeling and solution of the IES, including the power-gas IES [2], the power-heating IES, and the power-gas-heating IES [3], [4].

The techniques for solving the load flow of an IES can be generally classified into the decomposed method and the integrated method. In our previous work [5], the decomposed solution to the hybrid power-water flow problem of the power-water IES (PW-IES) is proposed and this work focuses on the integrated solution.

The integrated solution solves all equations of various en-

ergy vectors simultaneously and the most widely-used method is the Newton-Raphson (N-R) method [2]–[3], where the unified Jacobian matrix is developed and updated during the iterations. Reference [4] proposes a fast decoupled multi-energy flow calculation method, where the Jacobian matrix is replaced by a diagonal and constant Jacobian matrix. Many studies have demonstrated the good applicability of the N-R method to the load flow solution of the IES. However, when applying the N-R method to the PW-IES, numerical differences among Jacobian elements are found because the power equations are expressed in per-unit value while the hydraulic equations of the WDN are in actual value. These numerical differences may yield an ill-conditioned Jacobian matrix, which may result in the numerical stability problem and even the convergence problem of the N-R solution. Therefore, we propose to per-unitize the WDN to improve the integrated N-R solution of the hybrid power-water problem.

It is understood that a well-chosen per-unit scheme for power system analysis can simplify the modeling and reduce the computational cost. However, other energy vectors including WDN are usually modeled with actual values. Although per-unitizing the gas and heating networks are stated in [2] and [3], respectively, the base value selection in per-unitizing and the effects of per-unit modeling on the Jacobian matrix as well as the N-R solutions are not fully addressed.

The main contributions of this letter are suggested as follows: ① the per-unit scheme for the WDN with a focus on base value selection is proposed; ② the per-unitized hydraulic equations and the unified Jacobian matrix for the integrated N-R solution to the hybrid power-water flow problem are developed; and ③ the effects of per-unitized WDN on Jacobian matrix conditioning and the integrated N-R solution are discussed with both theoretical analysis and numerical studies.

II. HYBRID POWER-WATER FLOW MODELING OF PW-IES

A. Power Flow Equation

The AC power flow equations are used to model the power system as:

$$P_{Gi} - P_{Li} = V_i \sum_{j \in i} V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) \quad (1)$$

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$$Q_{Gi} - Q_{Li} = V_i \sum_{j \in i} V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) \quad (2)$$

where $j \in i$ represents the set of buses connecting to bus i ; V_i is the voltage magnitude of bus i ; θ_{ij} is the angle difference between bus i and bus j ; G_{ij} and B_{ij} are the real and imaginary parts of the element in the admittance matrix on bus i and bus j , respectively; P_{Gi} and Q_{Gi} are the active and reactive power generation on bus i , respectively; and P_{Li} and Q_{Li} are the active and reactive power loads on bus i , respectively.

B. Hydraulic Equation of WDN

A typical WDN consists of reservoirs, tanks, pumps, valves, pipes, and loads. The network can be represented by a set of nodes and edges. The set of nodes consists of the reservoir and tank nodes with known hydraulic heads or flows and other nodes with unknown hydraulic heads. The set of edges represents pipes, pumps, and valves with unknown flows. Valves can be treated as a special kind of pipes. Therefore, the only models of pipes and pumps will be presented below.

The steady-state characteristic of a pipe is mainly described by the hydraulic head loss across the pipe. q_{pk} is the flow rate of pipe k , which connects nodes i and j with hydraulic heads H_i and H_j , respectively. The hydraulic head loss ΔH_{ij} can be expressed as [6]:

$$\Delta H_{ij} = H_i - H_j = \gamma_{pk} q_{pk} \left| q_{pk} \right|^{n-1} \quad (3)$$

where the subscript p denotes pipes; n is the variable depending on the hydraulic head loss model; and γ_{pk} is the resistance coefficient of pipe k and can be expressed as:

$$\gamma_{pk} = \alpha L_{pk} / (C_{pk}^n D_{pk}^m) \quad (4)$$

where L_{pk} , D_{pk} , and C_{pk} are the length, diameter, and roughness coefficient of pipe k , respectively; and α and m are variables depending on the hydraulic head loss model. In this letter, we use the Hazen-Williams equation to describe the hydraulic head loss of pipes, where α , m , and n are 10.67, 4.871, and 1.852, respectively [6].

The widely-used variable-speed centrifugal pump is considered. q_{uk} is the flow rate of pump k with the from-node i and the to-node j . The hydraulic head gain of the pump is then modeled as [5]:

$$\Delta H_{uk} = H_j - H_i = \omega_{uk}^2 H_{0,uk} - r_{uk} \omega_{uk}^{2-m_{uk}} q_{uk}^{m_{uk}} \quad (5)$$

where the subscript u denotes pumps; $H_{0,uk}$, r_{uk} , and m_{uk} are the static head, internal resistance coefficient, and hydraulic head exponent of pump k , respectively; and ω_{uk} is the relative speed of pump k , i.e., the ratio of working speed to the nominal speed.

Equations (3) and (5) can be unified into the general form:

$$H_i - H_j = q_k f(q_k) \quad (6)$$

where q_k is the flow rate of pipe k or pump k connecting between node i and node j ; and $f(q_k)$ is defined as:

$$f(q_k) = \begin{cases} \gamma_{pk} |q_k|^{n-1} & \text{edge } k \text{ is a pipe} \\ -\omega_{uk}^2 H_{0,uk} q_k^{-1} + r_{uk} \omega_{uk}^{2-m_{uk}} q_k^{m_{uk}-1} & \text{edge } k \text{ is a pump} \end{cases} \quad (7)$$

Besides (6), the WDN is also governed by the continuity equations, which represent mass conservation requiring that the total flow into a node equals the flow out:

$$q_{wi} = \sum_{k \in E^-(i)} q_k - \sum_{k \in E^+(i)} q_k \quad (8)$$

where $E^+(i)$ and $E^-(i)$ are the sets of edges (including pipes and pumps) entering and leaving node i , respectively; and q_{wi} is the net flow rate injected into node i .

For a WDN with n_e edges connecting to n_n unknown-head-nodes and n_0 known-head-nodes, we define the incidence matrix A_w relating edges to nodes as:

$$A_w(i, j) = \begin{cases} -1 & \text{edge } i \text{ leaves node } j \\ 1 & \text{edge } i \text{ enters node } j \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Partitioning A_w into the following two submatrices, we can obtain:

$$A_w = [A_{12} \ A_{10}] \quad (10)$$

where A_{12} is the $n_e \times n_n$ matrix relating the edges to the nodes with unknown hydraulic heads; and A_{10} is the $n_e \times n_0$ matrix relating the edges to the nodes with known hydraulic heads.

The hydraulic equations of the WDN can then be written in the matrix form as:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{12}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \mathbf{H} \end{bmatrix} = \begin{bmatrix} -A_{10} \mathbf{H}_0 \\ -\mathbf{q}_w \end{bmatrix} \quad (11)$$

where $\mathbf{q}$ is the $n_e \times 1$ vector of edge flow rates; $\mathbf{H}$ is the $n_n \times 1$ vector of hydraulic heads to be determined; $\mathbf{H}_0$ is the $n_0 \times 1$ vector of known hydraulic heads; $\mathbf{q}_w$ is the $n_e \times 1$ vector of known nodal injected flow rates; and A_{11} is the $n_e \times n_e$ diagonal matrix with the elements as:

$$A_{11}(k, k) = f(q_k) \quad (12)$$

where $f(q_k)$ is the general function of pipe k or pump k .

C. Coupling Between Power System and WDN

Electricity-driven pumps are the main coupling components between the power system and the WDN. For pump k with ΔH_{uk} and q_{uk} , the power consumption P_{uk} is [5]:

$$P_{uk} = \rho g \Delta H_{uk} q_{uk} / \eta_{uk} \quad (13)$$

where ρ is the water density; g is the gravitational acceleration constant; and η_{uk} is the efficiency of pump k .

It should be noted that the power factors of pumps are assumed to be constant and the reactive power consumption Q_{uk} of pump k can be determined accordingly. For the pump connecting to node i of the power system, the active and reactive power consumption are incorporated into P_{Li} and Q_{Li} in (1) and (2), respectively.

III. PER-UNIT MODELING OF HYDRAULIC EQUATIONS

A. Illustrative Examples of Numerical Differences

The nominal voltages are usually selected as base voltages, and therefore, the per-unit voltages are near 1.0 under normal operating conditions. However, for the WDN, a wide

numerical range of parameters or variables can be found because of the actual values used. For example, the hydraulic heads of the 13-node WDN sample system are found to be in the range of [0, 670]m. Other examples are pipe parameters. The pipe lengths and diameters for a typical WDN generally lie in [500, 1000]m and [100, 400]mm [7], respectively, and a much wider range of [91, 1.56×10^5] for the resistance coefficients of pipes can thus be obtained with the Hazen-Williams equation.

B. Per-unitizing Hydraulic Equations

Similar to the per-unit scheme of the power system, the per-unit value of any quantity of a WDN is defined as the ratio of its actual value to the base value. Therefore, the most important thing for per-unitizing a WDN is the method to select the base values.

To develop the per-unit scheme for WDN analysis, we first make a comparison between the variables of the power system and the WDN, as shown in Table I, where the equivalent power S_w is defined from the power consumption of pumps in (13).

TABLE I
COMPARISONS BETWEEN VARIABLES OF POWER SYSTEM AND WDN

Scenario	Parameter	Variable
Power system	Voltage	V
	Voltage drop	ΔV
	Current	I
	Impedance	$Z = \Delta V / I$
	Apparent power	$S = VI$
WDN	Hydraulic head	H
	Head loss	ΔH
	Flow rate	q
	Resistance coefficient	$\gamma = \Delta H / q^n$
	Equivalent power	$S_w = \rho g \Delta H q$

To per-unit the power system equations, base power S_B and base voltage V_B are usually specified, and base current and base impedance are then determined to follow the circuit laws or relations between system variables. To define the per-unit scheme for the WDN analysis, the base flow rate q_B and base hydraulic head H_B are selected independently, and the base values of resistance coefficient γ_B and S_{WB} are determined as:

$$\gamma_B = H_B / q_B^n \quad (14)$$

$$S_{WB} = \rho g H_B q_B \quad (15)$$

where the subscript B denotes the base value of each variable.

The resistance coefficient r depends on the individual pump, then the base value for pump k is defined as:

$$r_{uB,k} = H_B / q_B^{m_{uk}} \quad (16)$$

With the base values, (11) and (13) can be per-unitized, respectively, as:

$$\begin{bmatrix} A_{11}^* & A_{12}^* \\ A_{12}^{*T} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{q}^* \\ \mathbf{H}^* \end{bmatrix} = \begin{bmatrix} -A_{10}^* \mathbf{H}_0^* \\ -\mathbf{q}_w^* \end{bmatrix} \quad (17)$$

$$P_{uk}^* = \Delta H_{uk}^* q_{uk}^* / \eta_{uk} \quad (18)$$

where the superscript * denotes the per-unit value; and A_{11}^* is defined as:

$$A_{11}^*(k, k) = f(q_k^*) = \begin{cases} \gamma_{pk}^* |q_k^*|^{n-1} & \text{edge } k \text{ is a pipe} \\ -\omega_{uk}^2 H_{0,uk}^* (q_k^*)^{-1} + r_{uk}^* \omega_{uk}^{2-m_{uk}} (q_k^*)^{m_{uk}-1} & \text{edge } k \text{ is a pump} \end{cases} \quad (19)$$

Note that when incorporating pump powers into (1), the base power of P_{uk} should be converted to the base power of the power system P_{uk}^* , which leads to:

$$P_{uk}^* = P_{uk} S_{WB} / S_B \quad (20)$$

Another point worth noting is that the hydraulic equations of a WDN should be per-unitized on a common q_B for all pipes or pumps and a common H_B for all nodes throughout the network.

C. Base Value Selection of Flow Rate and Hydraulic Head

Similar to the selection of S_B and V_B of the power system, both q_B and H_B can be selected quite arbitrarily. However, a well-chosen q_B and H_B can help simplify the calculation and make the per-unit values more informative. In this letter, we suggest to select q_B and H_B with a reference to the characteristic parameters of pipes and pumps, respectively.

According to [7], typical ranges of pipe diameters and the economic flow rates are listed in Table II.

TABLE II
TYPICAL RANGES OF PIPE DIAMETERS AND ECONOMICAL FLOW RATES

Diameter (mm)	Flow rate (m ³ /s)
100-400	0.005-0.113
>400	0.113-0.176

In the light of Table II, the following steps for selecting q_B are suggested: ① calculate the average diameter of all the pipes roughly, and ② set q_B to be the typical economic flow rate. For example, for a WDN with an average diameter less than 400 mm, q_B is suggested to be 0.01-0.1 m³/s.

As for H_B , the static heads of all the pumps shown in (5) are used by setting $H_B = \max(H_{0,u1}, H_{0,u2}, \dots, H_{0,um})$, where m is the total number of pumps.

IV. SOLUTION TO HYBRID POWER-WATER FLOW

A. N-R Method

Equations (1), (2), (11), and (13) constitute the hybrid power-water flow problem, where the power equations are expressed in per-unit value while the hydraulic equations are in actual value. This hybrid flow problem $F(\cdot)$ can be written in the general form as:

$$\mathbf{F}(\mathbf{X}) = \mathbf{0} \quad (21)$$

where $X=[\theta \ V \ q \ H]^T$ contains the unknown variables of the hybrid power-water flow problem, and θ and V are the vectors of the magnitudes and angles of unknown voltage in the power system, respectively, and q and H are the vectors of flow rates and hydraulic heads of unknowns in the WDN, respectively.

Applying the N-R method to (21), the iteration scheme is:

$$X^{(k+1)} = X^{(k)} - [J^{-1}]^{(k)} F^{(k)} \quad (22)$$

where the superscripts k and $k+1$ denote the current iteration and the $(k+1)^{\text{th}}$ iteration; and J is the Jacobian matrix calculated by:

$$J = \begin{bmatrix} J_{ee} & J_{ew} \\ \mathbf{0} & J_{ww} \end{bmatrix} = \begin{bmatrix} \frac{\partial P}{\partial \theta} & \frac{\partial P}{\partial V} & \frac{\partial P}{\partial q} & \frac{\partial P}{\partial H} \\ \frac{\partial Q}{\partial \theta} & \frac{\partial Q}{\partial V} & \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial H} \\ \mathbf{0} & \mathbf{0} & \mathbf{D} & \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{12}^T & \mathbf{0} \end{bmatrix} \quad (23)$$

where J_{ee} and J_{ww} are the Jacobian matrices of the power system and the WDN, respectively; J_{ew} is the derivative matrix of the power equations with respect to flow rates and hydraulic head; P and Q are the active and reactive power functions, respectively; and D is a diagonal matrix calculated by:

$$D(k, k) = \begin{cases} n\gamma_{pk}|q_k|^{n-1} & \text{edge } k \text{ is a pipe} \\ m_{uk}r_{uk}\omega_{uk}^{2-m_{uk}}(q_k)^{m_{uk}-1} & \text{edge } k \text{ is a pump} \end{cases} \quad (24)$$

Note that J_{ew} in (23) is non-zero due to the power consumption of pumps. More specifically, for bus i supplying the power to the pump k connecting nodes m and n , both P_{Li} in (1) and Q_{Li} in (2) are the functions of the unknown hydraulic heads H_m and H_n , and the unknown flow rate q_k of the pump.

The N-R iteration in (22) proceeds until the maximum magnitude of $F(X)$ is less than the specified tolerance or the iteration count exceeds the maximum number of iterations.

B. Jacobian Matrix in Per-unit

As mentioned above, when expressing hydraulic equations in actual values, there are significant numerical differences between the quantities of the two networks. Such significant numerical differences can also be found in Jacobian elements. More importantly, these differences may yield a great condition number of J , and thus lead to the numerical stability problem and even the convergence problem. Here, we provide a simple illustration of this statement.

λ is the eigenvalues of J , and the values of λ satisfy:

$$|\lambda E - J| = |\lambda_e E_e - J_{ee}| |\lambda_w E_w - J_{ww}| = 0 \quad (25)$$

where E , E_e , and E_w are the identity matrices; and λ_e and λ_w are the eigenvalues of J_{ee} and J_{ww} , respectively.

Since the proposed per-unit scheme is only used to deal with the hydraulic equations, nothing changes in λ_e and therefore only λ_w will be discussed below. J_{ww} can be written in (26) and $\lambda_{w1} \geq \lambda_{w2} \geq \dots \geq \lambda_{wn}$, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$, and $\tau_1 \geq \tau_2 \geq \dots \geq \tau_n$ are the eigenvalues of J_{ww} , B , and A , respectively, where $n = n_e + n_n$.

$$J_{ww} = B + A = \begin{bmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{0} & \mathbf{A}_{12} \\ \mathbf{A}_{12}^T & \mathbf{0} \end{bmatrix} \quad (26)$$

According to the min-max theorem and interlacing theorem of eigenvalues [8], we have:

$$\begin{cases} \sigma_i + \tau_n \leq \lambda_{wi} \leq \sigma_i + \tau_1 \\ \lambda_{w1} \geq \sigma_1 \geq \lambda_{w2} \geq \sigma_2 \geq \dots \geq \sigma_n \geq \lambda_{wn} \end{cases} \quad i = 1, 2, \dots, n \quad (27)$$

Considering that ① D is a positive and diagonal matrix, which means $\sigma_1 = \max\{D(i, i)\}$ and $\sigma_n = 0$; and ② $A(i, j) = \{0, 1, -1\}$ ($i \neq j$), $A(i, i) = 0$ and $|A| = 0$ ($n_e > n_n$), which means τ_n is a small negative number while τ_1 is a small positive number. We can conclude from (27) that a significant numerical difference in the elements of D will yield a great range of λ_{wi} . The 2-norm condition number of J_{ww} , denoted by $\kappa(J_{ww})$, is proven to be $\max\{|\lambda_{wi}|\}/\min\{|\lambda_{wi}|\}$. One can decrease $\kappa(J_{ww})$ by reducing the numerical differences among $D(i, i)$, which is the motivation of this letter to per-unitize the WDN.

Expressing the hydraulic equations in per-unit as in (17) and (18), the hybrid power-water flow problem in per-unit and its N-R iteration scheme, which have similar forms as (21) and (22), can be obtained. The Jacobian matrix in per-unit can be written as:

$$J^* = \begin{bmatrix} J_{ee} & J_{ew}^* \\ \mathbf{0} & J_{ww}^* \end{bmatrix} = \begin{bmatrix} \frac{\partial P}{\partial \theta} & \frac{\partial P}{\partial V} & q_B \frac{\partial P}{\partial q} & H_B \frac{\partial P}{\partial H} \\ \frac{\partial Q}{\partial \theta} & \frac{\partial Q}{\partial V} & q_B \frac{\partial Q}{\partial q} & H_B \frac{\partial Q}{\partial H} \\ \mathbf{0} & \mathbf{0} & \mathbf{D}^* & \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{12}^T & \mathbf{0} \end{bmatrix} \quad (28)$$

where D^* is the per-unit counterpart of D , which is a diagonal matrix given by:

$$D^*(k, k) = \begin{cases} n\gamma_{pk}^*|q_{pk}^*|^{n-1} & \text{edge } k \text{ is a pipe} \\ m_{uk}^*r_{uk}^*\omega_{uk}^{2-m_{uk}^*}(q_{uk}^*)^{m_{uk}^*-1} & \text{edge } k \text{ is a pump} \end{cases} \quad (29)$$

In the case studies, we will verify numerically that the condition number of the Jacobian matrix can be reduced significantly with the proposed per-unit scheme.

C. Initialization of Unknowns

For the N-R method, the initialization should be done carefully. A common practice for initializing power flow is the flat start, i.e., set all unknown voltage $V_i = 1$ and the unknown angle $\theta_i = 0$. This flat start can be extended to a per-unitized WDN by setting the per-unit value of all unknown flow rates $q_i = 1$ and $H_i = 1$. Due to the special consideration in selecting q_B and H_B as presented above, this initialization is usually proven to be a good start.

V. CASE STUDIES

A. Test System Description

Two test systems are studied below to illustrate the advantage of the proposed per-unit scheme in the analysis of hybrid power-water flow. The first system is composed of the modified IEEE 33-bus system incorporating two photovoltaic sources and a 13-node WDN, which is referred to as

IEEE 33-W13 below. The second system, denoted by IEEE 33-W27, is composed of the same IEEE 33-bus system and a 27-node WDN.

B. Numerical Differences in Jacobian Matrix

In Table III, we compare the maximum or minimum Jacobian elements in actual and per-unit values. It can be observed that the value range of Jacobian elements is significantly reduced by per-unitizing.

TABLE III
MAX/MIN OF JACOBIAN ELEMENTS IN ACTUAL AND PER-UNIT VALUES

Value	IEEE 33-W13		IEEE 33-W27	
	Max	Min	Max	Min
Actual value	14632.304	-21.494	576.046	-21.494
Per-unit value	21.494	-21.494	21.494	-21.494

Figure 1 shows the colormaps of the Jacobian matrix of the IEEE 33-W27 system, where the WDN is expressed in actual and per-unit values with the proposed per-unit scheme. The dimensions of $\mathbf{J}_{cc}$, $\mathbf{J}_{cw}$, and $\mathbf{J}_{ww}$ are 62×62 , 62×60 , and 60×60 , respectively, and a and b represent the coordinates of the elements in the Jacobian matrix.

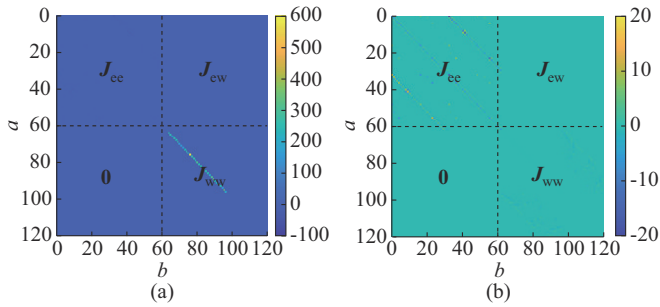


Fig. 1. Colormaps of Jacobian matrix of IEEE 33-W27 system. (a) Actual value. (b) Per-unit value.

In Fig. 1(a), there is a line of bright dots in the part of $\mathbf{J}_{ww}$ while the four parts of Fig. 1(b) are almost in the same color. The bright dots represent the diagonal elements of $\mathbf{D}$ defined in (23) and the brightness indicates that the elements of $\mathbf{D}$ are quite different from other elements. This is because $\mathbf{D}$ is determined by the hydraulic parameters of pipes and pumps while other elements of $\mathbf{J}_{ww}$ are 0, 1, and -1. Moreover, since the power equations are expressed in per-unit value, the elements of both $\mathbf{J}_{cc}$ and $\mathbf{J}_{cw}$ are quite small compared with the elements of $\mathbf{D}$ in Fig. 1(a).

Both Table III and Fig. 1 verify that the proposed per-unit scheme can reduce the numerical difference among Jacobian elements significantly.

C. Condition Number of Jacobian Matrix

Table IV compares the 2-norm condition numbers of the Jacobian matrix in actual and per-unit values for the two test systems. The comparison shows that the condition numbers of the per-unit Jacobian matrix for both systems decrease significantly. Note that the condition numbers of the per-unit Jacobian matrix for the two test systems are the same. This is

because the maximum and minimum singular values of the per-unit Jacobian matrix for the two systems are both determined by the IEEE 33-bus system.

TABLE IV
2-NORM CONDITION NUMBERS OF JACOBIAN MATRIX

Value	IEEE 33-W13	IEEE 33-W27
Actual value	4.4932×10^8	4.7478×10^6
Per-unit value	2.4641×10^3	2.4641×10^3

D. Analysis of Ill-conditioned System

For hydraulic analysis, a large variation in pipe lengths, diameters, and roughness may result in an ill-conditioned system [6]. To investigate the impact of pipe parameters on Jacobian matrix conditioning and the proposed per-unit scheme to deal with the possible ill-conditioning, the following three cases of the IEEE 33-W27 are presented.

Case 1: the original parameters of the WDN are considered.

Case 2: the length of pipe W1-W2 increases to three times the original length.

Case 3: the diameter of pipe W1-W2 reduces to one third of the original diameter.

Table V compares the 2-norm condition number of the Jacobian matrix $\kappa(\mathbf{J})$, the N-R iterations k , and the computational time T in solving the three cases. All computations are performed on a laptop, equipped with an Intel Core i5 CPU @ 1.8 GHz and 8 GB of RAM. Note that the same N-R starting point or actual tolerances for actual and per-unit values are considered. For example, the tolerance of 10^{-10} for hydraulic heads in per-unit value is converted to the actual tolerance by multiplying H_B . Moreover, the singularity of the Jacobian matrix is also considered as the third N-R termination condition besides the tolerance and the maximum iterations mentioned above.

TABLE V
CONVERGENCE COMPARISON OF THREE CASES

Case	Actual value			Per-unit value		
	$\kappa(\mathbf{J})$	k	T (ms)	$\kappa(\mathbf{J}^*)$	k	T (ms)
1	4.7478×10^6	5	7	2.4641×10^3	5	6.5
2	5.3035×10^6	5	7	2.4641×10^3	5	7.0
3	9.8541×10^{14}	-	-	6.9513×10^6	6	8.1

Note: “-” represents non-convergent.

In Table V, there are obvious differences among $\kappa(\mathbf{J})$ of the three cases, and the most obvious difference lies in case 3. Compared with $\kappa(\mathbf{J})$, the values of $\kappa(\mathbf{J}^*)$ are significantly reduced by the proposed per-unit scheme. Another observation is that both iteration number and computation time for the solved cases are almost the same.

Case 3 in actual value is the only unsolved case. The eigenvalue distributions of the Jacobian matrix of Case 3 are shown in Fig. 2. As can be observed, the eigenvalue distribution of $\mathbf{J}^*$ is much more concentrated than that of $\mathbf{J}$. Besides, there is a very large eigenvalue of $\mathbf{J}$, which results in an ex-

tremely large condition number and a convergence problem.

Both Table V and Fig. 2 verify the better convergence of the proposed per-unit scheme and its ability to relieve the ill-conditioning of the Jacobian matrix.

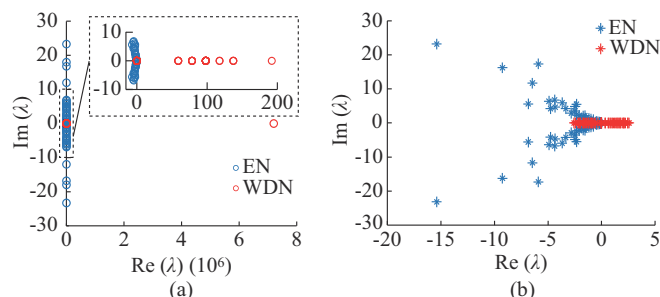


Fig. 2. Eigenvalue distribution. (a) J . (b) $\bar{J}$.

VI. CONCLUSION

This letter proposes a per-unit scheme for the WDN to improve the integrated N-R solution for the hybrid power-water flow problem of the PW-IES. The effects of per-unitizing on the Jacobian matrix conditioning are analyzed theoretically and the improvements in N-R solution are verified numerically through two test systems.

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